



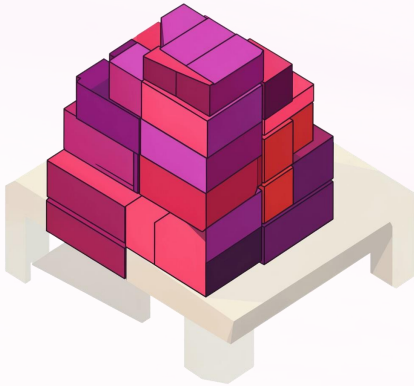
Python Modules & Libraries

The building blocks that make Python powerful, efficient, and beloved by millions of developers worldwide.

PYTHON FUNDAMENTALS

Made with GAMMA

What Are Modules & Libraries?



Module

*A **single file** containing Python definitions — functions, classes, and variables.*

Library

*A **collection of modules** providing pre-written code for specific tasks.*

Why It Matters

*Organise code into **logical, reusable units** — write once, use everywhere.*

Why Modules & Libraries Are Crucial



Batteries Included

*Extensive functionality **out of the box***



Code Reusability

*Import and reuse across **multiple projects***



Modularity

*Break complex programs into **manageable parts***



Maintainability

*Isolated changes, **simpler updates***



Namespace Management

*Private namespaces, **no naming conflicts***

Practical Uses: Extending Python's Power



math

Advanced maths —
`math.sqrt()`, trigonometry



datetime

Handle dates, times, and
calculations



os

OS interactions —
`os.getcwd()`, file paths



json

Parse and encode JSON for
web apps



NumPy & Pandas

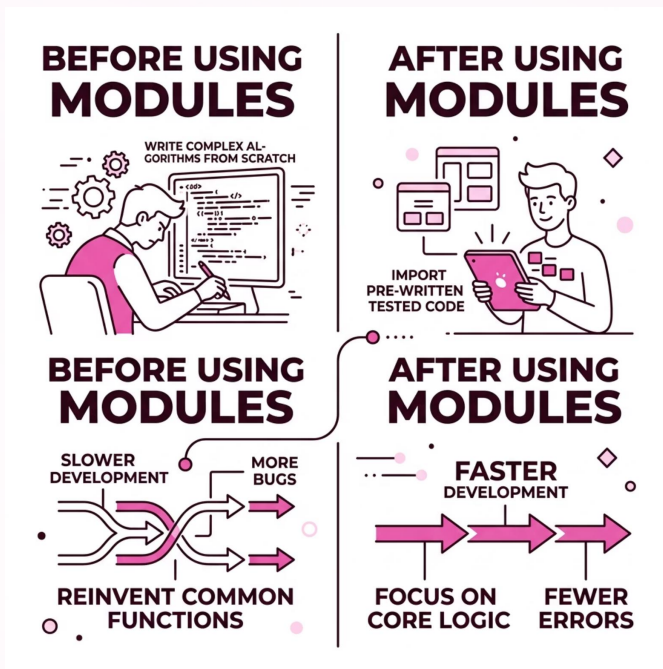
Numerical ops and data
analysis



Django

Full-stack web development
framework

How Modules Reduce Your Workload



- Pre-written Functionality
Import tested code instead of writing from scratch
- Faster Development
Leverage optimised, battle-tested libraries
- Focus on Core Logic
Concentrate on what makes your app unique

Example: `import math; result = math.sqrt(16)` — one line instead of complex numerical methods.

Real-World Impact



Data Analysis

Pandas & NumPy handle **large datasets** efficiently



Web Development

Django & Flask streamline **dynamic websites & APIs**



Machine Learning

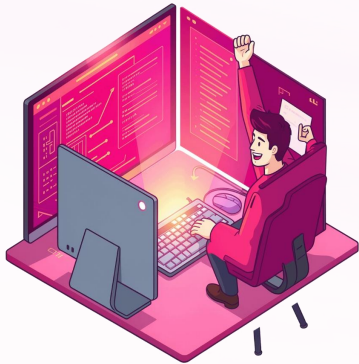
TensorFlow & PyTorch power **advanced AI applications**



System Admin

os & sys modules script **system-level tasks**

Conclusion: Unlock Python's True Potential



Fundamental to Python's Success

Modules drive Python's versatility and adoption

Don't Reinvent the Wheel

Build sophisticated apps efficiently with existing tools

Your Coding Journey

*Simpler, faster, and **more powerful** with modules*



Key Takeaway: Master modules to write smarter, not harder.